

Tiny-vGamba: Distilling Large Vision-(Language) Knowledge from CLIP into a Lightweight vGamba Network

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Abstract

Deploying high-performing visual models on resource-constrained edge devices remains a challenge due to the computational demands of architectures like Vision Transformers (ViT) and large vision-language models such as CLIP (Contrastive Language-Image Pre-training). In this work, we propose Tiny-vGamba, a lightweight visual recognition backbone designed for efficient deployment in low-latency settings while efficiently modeling long-range dependencies. Building upon the original vGamba architecture, which combines Mamba-based state-space modeling with attention mechanisms, we simplify the design by removing memory-intensive components such as global self-attention and fusion gates, retaining a streamlined Mamba Gamba-Cell for efficient long-range dependency modeling. To further enhance generalization, we introduce a knowledge distillation framework that transfers visual-semantic knowledge from a frozen CLIP (ViT-B/32) teacher to the Tiny-vGamba student. This includes logit and feature-level supervision, mean-variance alignment, and lightweight channel attention to guide the student's learning. Experimental results on lightweight classification, detection, and segmentation tasks enhance performance, outperforming several lightweight baselines, which are well-suited for real-time edge applications. https://github.com/newralab-ai/tiny_vGamba

1. Introduction

Modern visual recognition tasks increasingly rely on large-scale vision-language models such as CLIP (Contrastive Language-Image Pre-training) [21], which learns powerful semantic representations from hundreds of millions of

image-text pairs [11]. These models have achieved remarkable zero-shot generalization and capture long-range dependencies, largely attributed to their Transformer-based architectures and large training corpora. However, their high computational complexity, significant memory footprint, and latency make them impractical for edge deployment scenarios such as IoT, robotics, or autonomous navigation. A more viable approach for edge deployment is to transfer the knowledge of such large models into lightweight students through a teacher-student distillation framework. Figure 1 illustrates the improvement in generalization and class-wise accuracy after distilling visual knowledge from CLIP (ViT-B/32) into our Tiny-vGamba model.

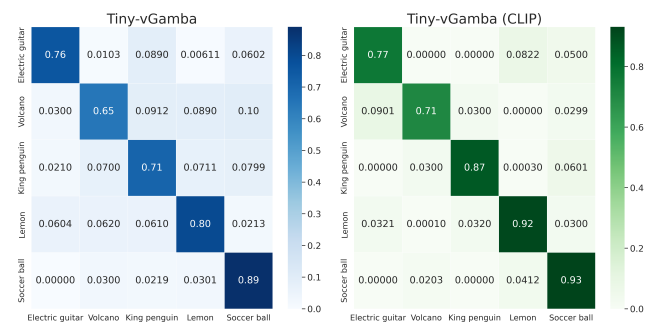


Figure 1. Confusion matrices for Tiny-vGamba and Tiny-vGamba (CLIP-distilled). We randomly selected 5 classes from ImageNet-1K: *electric guitar*, *volcano*, *king penguin*, *Lemon*, and *soccer ball*. The confusion matrix illustrates that after applying our proposed CLIP-based visual knowledge distillation framework, Tiny-vGamba (CLIP) shows notable improvements in generalization and class-wise accuracy for visually ambiguous categories.

To this trend, hybrid architectures like vGamba [5] try to balance efficiency and expressiveness by integrating Mamba [4], a Selective State Space Model (SSM) with

attention-based mechanisms in a Gamba Bottleneck design. While vGamba reduces inference costs compared to vanilla Vision Transformer (ViT) [2], its use of parallel Mamba and Multi-Head Self-Attention (MHSA) layers still imposes synchronization overhead and quadratic complexity, limiting its deployment on low-power edge devices. While lightweight backbones like EfficientVMamba-T [10], DeiTi [15], QuadMamba-Li [19], and ResNet-18 [7] are efficient in resource-constrained settings, they have limited generalization capacity compared to large-scale models like CLIP.

In this work, we propose Tiny-vGamba, a compact and edge-friendly visual backbone customized for efficient long-range dependency modeling. Our key insight is to remove the heavy *global* MHSA and fusion modules in the original Gamba bottleneck, retaining only the efficient Mamba-based Gamba-Cell. This design choice removes module-level synchronization and reduces complexity, enabling real-time inference on resource-constrained hardware. To enhance its generalization, we introduce a novel distillation framework that transfers knowledge from the CLIP (ViT-B/32) visual encoder to our Tiny-vGamba student. The distillation process leverages both logit-level KL divergence and feature-level mean-variance alignment, enhanced by a lightweight channel attention mechanism for adaptivity. This enables Tiny-vGamba to inherit CLIP's generalization capabilities while maintaining a minimal computational footprint.

The contributions of this work are summarized below:

- We propose Tiny-vGamba, a lightweight hybrid vision backbone model designed to enhance long-range dependencies in complex images using Mamba and CNN.
- We propose an efficient distillation framework that transfers large visual knowledge to Tiny-vGamba, reducing dependence on large labeled datasets.
- Experiments on classification, detection, and segmentation tasks show that Tiny-vGamba outperforms some existing lightweight models due to its efficient global context modeling and the proposed knowledge distillation framework.

2. Related Work

Vision-language models like CLIP [11] excel at aligning visual and textual modalities, enabling strong generalization, but are too large for edge deployment. ViT [2] and hybrid attention-based models effectively capture global context but suffer from quadratic complexity [6]. To mitigate this challenge, approaches such as sparse attention [1], lightweight architectures [10, 17–19, 23], and convolution-attention hybrids [8] have been proposed. Recently, SSM models like Mamba [4] offer linear scalability and efficient long-range modeling. However, these models do not match the strong generalization capabilities of CLIP [11].

To address the performance-efficiency trade-off, knowledge distillation has become critical. Traditional methods have evolved into vision-language distillation pipelines, enabling compact models to inherit CLIP's semantic understanding. Nonetheless, lightweight architectures often struggle to capture complex visual-semantic relationships, and distillation may fail to bridge structural mismatches between teacher and student models. This motivates the need for compact, expressive backbones paired with customized distillation strategies to enable efficient visual knowledge transfer.

3. Preliminaries

3.1. Contrastive Language–Image Pretraining

CLIP [11] is a *multi-modal dual-encoder* that aligns image–text pairs in a shared embedding space. Given N image–text pairs $\{(x_i, t_i)\}_{i=1}^N$, with encoders $f_v(\cdot)$ and $f_t(\cdot)$ for vision and text respectively, embeddings are ℓ_2 -normalized and compared via cosine similarity, computed in Eqn. 1.

$$s_{ij} = \frac{f_v(x_i) \cdot f_t(t_j)}{\|f_v(x_i)\| \|f_t(t_j)\|}. \quad (1)$$

The model is optimized using a symmetric cross-entropy loss over all N^2 image–text pairs in a batch, Eqn. 2.

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^N \left[-\log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^N \exp(s_{ij}/\tau)} - \log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^N \exp(s_{ji}/\tau)} \right], \quad (2)$$

where τ is a learnable temperature parameter that scales the logits. After training, CLIP supports *zero-shot inference* by encoding a set of prompts into text embeddings. For a given test image x , its label is predicted by comparing its visual embedding to the set of C text embeddings, as in Eqn. 3.

$$\hat{y} = \arg \max_k \frac{f_v(x) \cdot f_t(t_k)}{\|f_v(x)\| \|f_t(t_k)\|}. \quad (3)$$

CLIP [11] achieves strong zero-shot accuracy across tasks via modality alignment without supervision. Its ViT [2] backbone is often reused for generalization and prompt compatibility.

3.2. SSM

Represent temporal signals $x(t) \in \mathbb{R}$ using a hidden state $h(t) \in \mathbb{R}^N$, governed by a linear time-invariant system, as in 4.

$$\begin{aligned} \frac{d}{dt}h(t) &= Ah(t) + Bx(t), \\ y(t) &= Ch(t) + Dx(t), \end{aligned} \quad (4)$$

where $A \in \mathbb{R}^{N \times N}$ defines state dynamics, $B \in \mathbb{R}^{N \times 1}$ couples the input to the state, $C \in \mathbb{R}^{1 \times N}$ maps the state to output, and $D \in \mathbb{R}$ allows direct input-output interaction. To enable neural network integration, the continuous system is typically discretized using zero-order hold, assuming constant input over interval Δt :

$$\hat{A} = e^{A\Delta t}, \quad \hat{B} = \left(\int_0^{\Delta t} e^{A\tau} d\tau \right) B. \quad (5)$$

The resulting discrete-time recurrence becomes:

$$\begin{aligned} h_k &= \hat{A}h_{k-1} + \hat{B}x_k, \\ y_k &= Ch_k + Dx_k. \end{aligned} \quad (6)$$

While classical SSMs use fixed matrices A, B, C, D at inference, limiting their flexibility, modern variants such as Mamba introduce input-dependent parametrization, e.g., dynamic B, C , and step size, enabling better feature selection and long-range dependency modeling in high-dimensional sequences.

4. Method

Our goal is to develop a lightweight visual recognition backbone that retains the semantic strength and generalization capabilities of CLIP large-scale model while remaining suitable for deployment on edge devices. To achieve this, we propose Tiny-vGamba, a compact architecture that distills visual knowledge from a frozen CLIP (ViT-B/32) encoder. The method section consists of three main components: (1) adapting the Gamba bottleneck for low-latency inference, (2) transferring visual knowledge via feature and logit distillation, and (3) improving feature alignment through statistical matching and lightweight channel attention.

4.1. Problem Definition

Large-scale vision-language models like CLIP demonstrate strong generalization and semantic grounding by learning from over 400 million image/text pairs. However, their high computational cost and memory demands make them unsuitable for edge deployment. While lightweight visual backbones are more efficient, they often struggle to model long-range dependencies and global contexts due to their inherently shallow network architectures. In this work, we bridge this gap by proposing a lightweight, compact visual recognition backbone that retains the strong generalization and semantic understanding present in the CLIP model. Specifically, we distill knowledge from the visual encoder of a frozen CLIP model (ViT-B/32) into a lightweight, edge-friendly architecture. Our proposed Tiny-vGamba integrates convolutional layers with a Gambacell bottleneck, utilizing a Mamba-based state-space module.

This combines CNN efficiency for local feature extraction with lightweight long-range dependency and global context modeling via Mamba, enabling efficient and semantically aware inference custom for resource-constrained environments using only the visual knowledge learned by CLIP.

4.2. Adapting Gamba for Lightweight settings

The original vGamba [5] replaces ResNet [7] bottlenecks with Gamba bottlenecks, a hybrid module that combines Mamba [4], global MHSA, and a spatial gating mechanism to improve long-range dependency and global context modeling. However, these introduce overhead, limiting edge deployment. The global MHSA is memory intensive due to its quadratic complexity in input resolution, requiring large attention maps and intermediate storage. Additionally, the parallel arrangement of Mamba and MHSA modules can cause synchronization delays, as one module may wait for the other, increasing latency and reducing efficiency in real-time scenarios.

To address these challenges, we propose Tiny-vGamba, a streamlined variant optimized for low-latency, resource-constrained environments. We remove the global MHSA and fusion gating mechanism while retaining the Mamba-based module, which offers approximately $5\times$ efficiency over ViT for sequence modeling. Our lightweight Tiny Gamba Cell processes input feature maps $X \in \mathbb{R}^{B \times C \times H \times W}$ by reshaping them into sequence form $X_{\text{seq}} \in \mathbb{R}^{B \times (HW) \times C}$, adding 2D relative positional encodings $P = R_h + R_w$, and passing the positional-aware sequence through the Mamba block: $X_{\text{mamba}} = \text{Mamba}(X_{\text{seq}})$. The result is then passed through the Gated Spatial Context module, activated using GLU, and reshaped back to the spatial format: $X_{\text{out}} = \text{GLU}(\text{GSC}(X_{\text{mamba}})) \in \mathbb{R}^{B \times C \times H \times W}$. For architectural simplicity, we build Tiny-vGamba on ResNet-18 backbones, replacing its final convolutional blocks (3×3 Conv2D) with Tiny Gamba Cell. This design preserves local feature learning while enabling efficient long-range and global context modeling, ideal for real-time edge deployment.

4.3. CLIP (ViT-B/32) Knowledge Distillation

To transfer vision knowledge from the large-scale CLIP teacher model to our lightweight Tiny-vGamba student model, we propose a knowledge distillation framework. Specifically, we utilized a dual-component loss function that combines the Kullback–Leibler (KL) divergence between the softened outputs of the student and teacher models with the standard cross-entropy (CE) loss for supervision. In addition, we introduce a lightweight and learnable channel attention layer in the output of the visual features of the teacher to improve channel-wise adaptivity before distillation into the student, which is essential for long-range dependencies.

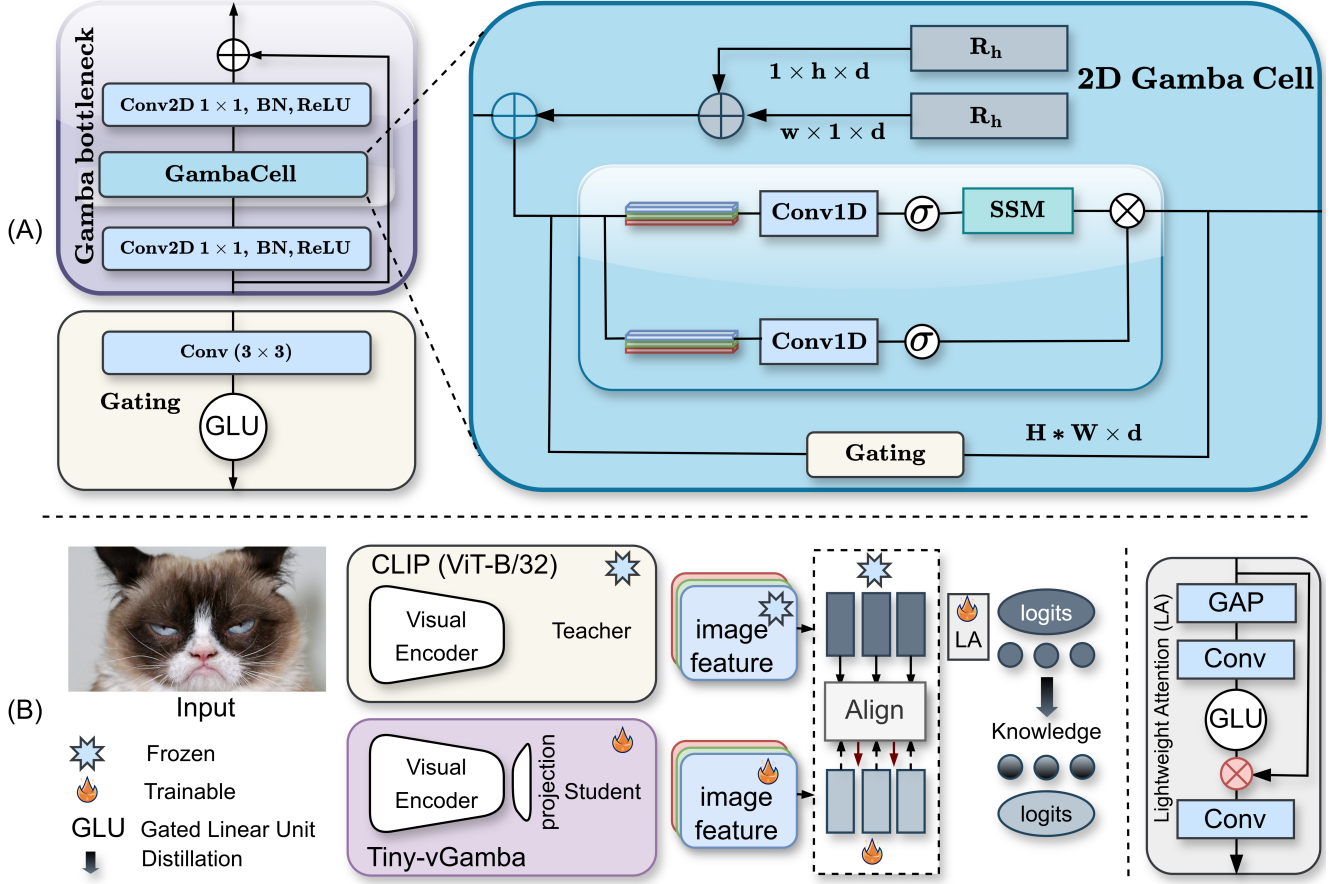


Figure 2. Overview of our proposed knowledge distillation strategy for the Tiny-vGamba model: (a) A Gamba Bottleneck adapted for low-resource edge devices; which includes the 2D positional embeddings, a mamba block and a gated spatial context (b) Tiny-vGamba (student) uses only the visual encoder of the CLIP (ViT-B/32) model as teacher. The framework introduced a lightweight attention (LA) and a feature-level mean-variance alignment.

The teacher model uses the visual encoder of CLIP (ViT-B/32), loaded from local checkpoints and not updated during training. For each input image, we extract normalized image embeddings from the teacher. In parallel, the Tiny-vGamba (student) model processes the same input to produce logits, which are also normalized. The total training loss is computed as in Eqn. 7.

$$\mathcal{L}_{\text{total}} = \alpha \cdot \text{KL} \left(\text{softmax} \left(\frac{s}{T} \right), \text{softmax} \left(\frac{t}{T} \right) \right) + (1 - \alpha) \cdot \text{CE}(s, y) \quad (7)$$

where s and t are the student and teacher logits respectively, y is the ground-truth label, T is the temperature parameter, and α is a balancing factor.

We implement this using a custom `Distillation Loss` class with temperature $T = 1.0$ and $\alpha = 0.5$. The student model is trained using the Adam optimizer over multiple epochs. The best-performing student checkpoint is selected based on test accuracy. This approach enables

Tiny-vGamba to learn CLIP (ViT-B/32) large visual knowledge while maintaining the computational efficiency necessary for deployment on edge devices.

4.3.1. Feature Alignment via Mean-Variance Matching

We align Tiny-vGamba (student) image features with the CLIP (ViT-B/32) by matching their mean and variance statistics. Let $\mathbf{u}_s$ and $\mathbf{u}_t$ be the student and teacher features, respectively. A learnable projector $P(\cdot)$ maps student features to the teacher's feature space. The means and variances are computed as in Eqn. 8.

$$\begin{aligned} \mu_s &= \frac{1}{N} \sum P(\mathbf{u}_s), \quad \mu_t = \frac{1}{N} \sum \mathbf{u}_t, \\ \sigma_s^2 &= \frac{1}{N} \sum (P(\mathbf{u}_s) - \mu_s)^2, \quad \sigma_t^2 = \frac{1}{N} \sum (\mathbf{u}_t - \mu_t)^2 \end{aligned} \quad (8)$$

The alignment loss is the sum of the L1 distances between these statistics, Eqn. 9.

$$L_{\text{align}} = \|\mu_s - \mu_t\|_1 + \|\sigma_s^2 - \sigma_t^2\|_1 \quad (9)$$

Minimizing L_{align} encourages the student model to mimic the teacher’s feature extraction effectively with minimal computational overhead.

4.3.2. Lightweight Attention (LA)

In addition, we introduce a lightweight, learnable channel attention layer on the teacher’s visual features to enhance channel-wise adaptivity before distillation. Given input features $x \in \mathbb{R}^{C \times H \times W}$, we first apply adaptive average pooling spatially to obtain $p \in \mathbb{R}^{C \times 1 \times 1}$. Then, a 1×1 convolution followed by a Gated Linear Unit (GLU) activation produces channel attention weights Z .

where $Z \in \mathbb{R}^{C \times 1 \times 1}$ recalibrates the input features via element-wise multiplication as in Eqn. 10.

$$\text{LA} = x \otimes Z \quad (10)$$

This process efficiently captures inter-channel dependencies, improving the teacher’s feature representation of long-range relationships before knowledge distillation.

5. Experiments

Datasets: We evaluate Tiny-vGamba across three downstream tasks with commonly used benchmarks. For image classification, we use ImageNet-1K, which contains 1.2M training images and 50K validation images across 1,000 classes. For semantic segmentation, we adopt ADE20K with 20K training and 2K validation images annotated over 150 categories. Object detection is evaluated on COCO2017, comprising 118K training and 5K validation images with 80 object classes. In addition, the AID dataset is used to test for generalization and long-range dependencies; it has 10K aerial images across 30 scene categories.

Implementation Details: The initial training setup for the student model followed the original vGamba configuration. For knowledge distillation, we used only the visual encoder of CLIP (ViT-B/32), keeping it frozen as the teacher. The student model (Tiny-vGamba) was aligned to match the input resolution and output embedding dimension of the teacher (224×224 resolution, 512-dimensional embedding). A temperature-scaled KL divergence loss was employed for distillation, using a temperature of 1.0 and an α value of 0.5 to balance the distillation loss and the ground truth loss. The best-performing student checkpoint was selected based on test accuracy. All experiments were conducted using PyTorch with mixed precision training on 8 NVIDIA Titan XP GPUs (12GB each).

5.1. Classification

Results. Table 1 presents a performance comparison of lightweight vision backbones on ImageNet-1K. The proposed *Tiny-vGamba* achieves a Top-1 accuracy of 73.8% with 14M parameters and 1.9G FLOPs, outperforming early

Table 1. Model performance comparison on ImageNet-1K.

Method	IMAGENET-1K			
	Size	Param (M)	FLOPs (G)	Top-1 (%)
ResNet-18 [7]	224 ²	11.8	1.8	69.8%
MobileOne-S1 [17]	224 ²	4.8	0.8	75.9%
QuadMamba-Li [19]	224 ²	5.4	0.8	74.2%
EfficientVMamba-T [10]	224 ²	6.1	1.0	76.5%
RegNetY-800M [12]	224 ²	6.0	0.8	76.3%
PVTv2-B0 [18]	224 ²	3.0	0.6	70.5%
MobileViT-XS [9]	224 ²	2.0	1.0	74.5%
DeiT-Ti [15]	224 ²	6.0	1.3	72.2%
ViM-Ti [23]	224 ²	7.0	1.5	73.1%
ViTAS-DeiT-A [14]	224 ²	6.0	1.3	75.5%
PVT-T [18]	224 ²	13.2	1.9	75.1%
Tiny-vGamba	224 ²	14.0	1.9	73.8%
*Tiny-vGamba	224 ²	14.0	1.9	76.7% (+2.9)

* Model after knowledge distillation.

convolutional baselines like ResNet-18 (69.8%) and compact transformers such as DeiT-Ti (72.2%) and ViM-Ti (73.1%). Notably, after distillation with CLIP (ViT-B16), *Tiny-vGamba* further boosts performance to 76.7%, marking a +2.9% gain. This result is competitive with more efficient backbones like EfficientVMamba-T (76.5%) and RegNetY-800M (76.3%), despite using slightly higher computation. These findings demonstrate that the proposed distillation framework effectively transfers knowledge from a larger model without increasing computational cost, making it well-suited for resource-constrained vision tasks.

5.2. Object Detection

Results. Table 2 presents a comparison of lightweight backbones for object detection (AP_b) and instance segmentation (AP_m) on the COCO val2017 dataset. The proposed Tiny-vGamba achieves 38.6 AP_b and 33.2 AP_m with 206M FLOPs and 31M parameters, outperforming classic baselines such as ResNet-18 (34.0 / 31.2), PVT-T (36.7 / 35.1), PoolFormer-S12 (37.3 / 34.6), EfficientFormer-L1 (37.9 / 35.4), and FastViT-SA12 (38.9 / 35.9). Applying our proposed distillation strategy further improves performance significantly: CLIP (ViT-32B) distilled Tiny-vGamba achieves 41.4 AP_b and 38.5 AP_m , surpassing EfficientVMamba-S (39.3 / 36.7), QuadMamba-Li (39.3 / –), and even more advanced models like SwiftFormer-L1 (41.2 / 38.1) and ViL-T (41.4 / 38.1). This highlights the effectiveness of our distillation framework in transferring knowledge from a larger visual model, enabling *Tiny-vGamba* to achieve state-of-the-art results in lightweight settings without increasing computational cost or memory,

making it ideal for real-world, resource-constrained detection and segmentation tasks.

Table 2. Comparison with lightweight models on object detection and instance segmentation on COCO val2017 datasets.

Backbone	FLOPs		Params		AP_b	AP_m
	(G)	(M)	(G)	(M)		
ViL-T [22]	26	145	41.4	38.1		
SwiftFormer-L1 [13]	–	–	41.2	38.1		
EfficientVMamba-S [10]	31	197	39.3	36.7		
QuadMamba-Li [19]	25	186	39.3	–		
FastViT-SA12 [16]	–	–	38.9	35.9		
EfficientFormer-L1 [8]	–	–	37.9	35.4		
PoolFormer-S12 [20]	–	–	37.3	34.6		
PVT-T [18]	32	208	36.7	35.1		
ResNet-18 [7]	31	207	34.0	31.2		
Tiny-vGamba	31	206	38.6	33.2		
*Tiny-vGamba	31	206	41.4	38.5		

* Model after knowledge distillation.

5.3. Segmentation

Results. Table 3 presents a comparison of lightweight backbones on the ADE20K val set for semantic segmentation. The proposed *Tiny-vGamba* achieves a competitive mIoU of 39.1% (SS) and 39.8% (MS), outperforming PoolFormer-S12 (37.2%) and FastViT-SA12 (38.0%). After applying our proposed distillation strategy, *Tiny-vGamba* achieves a significant performance boost, reaching 41.9% (SS) and 42.3% (MS), surpassing EfficientVMamba-T (39.3%) and even outperforming EfficientVMamba-S (42.1%) in multi-scale inference, with fewer parameters (-1M). These results demonstrate the effectiveness of the distillation process, allowing *Tiny-vGamba* to absorb rich semantic knowledge from a larger model and deliver strong segmentation performance, making it well-suited for resource-constrained deployment in real-world semantic understanding tasks.

5.4. Ablation

5.4.1. Effects of LA and feature alignment

Table 4 presents an ablation study evaluating the individual and combined effects of the two key components in our distillation framework: Feature Alignment (FA) and Lightweight Attention (LA). Starting from the baseline *Tiny-vGamba* without any distillation component, we observe a Top-1 accuracy of 73.8%. Introducing FA alone results in a gain of +0.7%, indicating that aligning feature statistics between the student and the CLIP teacher helps improve representation learning. Similarly, using LA alone yields a slightly higher gain of +0.9%, showing that enhancing the teacher’s feature expressiveness via channel at-

Table 3. Semantic segmentation results on ADE20K val set.

Backbone	ADE20K			
	Param (M)	FLOP (G)	mIoU (SS)	mIoU (MS)
EfficientVMamba-T [10]	14	230	38.9	39.3
DeiT-Ti [15]	11	–	39.2	–
ViM-Ti [23]	13	–	40.2	–
EfficientVMamba-S [10]	29	505	41.5	42.1
PoolFormer-S12 [20]	–	–	–	37.2
EfficientFormer-L1 [8]	–	–	–	38.9
FastViT-SA12 [16]	–	–	–	38.0
Tiny-vGamba	28	603	39.1	39.8
*Tiny-vGamba	28	603	41.9	42.3

* Model after knowledge distillation.

tention aids in better knowledge transfer. When both components are applied together, our full distillation approach achieves a significant improvement of +2.9%, raising the Top-1 accuracy to 76.7%. This confirms the complementary benefits of FA and LA, demonstrating their joint effectiveness in improving the student model’s performance on ImageNet-1K.

Table 4. Ablation study on the effects of Feature Alignment (FA) and Lightweight Attention (LA) in the distillation framework. Results are reported on IMAGENET1K.

Components	FA	LA	Top-1
Tiny-vGamba (Baseline)	✗	✗	73.8%
+ FA Only	✓	✗	74.5% (+0.7)
+ LA Only	✗	✓	74.7% (+0.9)
+ FA + LA (Ours)	✓	✓	76.7% (+2.9)

5.4.2. Effect of loss configurations

Table 5 shows the empirical ablation of combining cross-entropy (CE) loss with knowledge distillation (KD) loss for optimal accuracy. Using CE loss alone achieves a baseline accuracy of 74.2%, while using only KD loss results in a lower 72.8%. Introducing both losses with varying temperature and alpha values consistently improves performance. The best accuracy of 76.7% is obtained at a temperature of 1.0 and an alpha of 0.5, confirming this as the most effective balance between the CE and KD terms. Slight deviations in temperature or alpha from these values result in modest accuracy drops, indicating the method is fairly robust to hyperparameter variations.

Overall, these findings highlight the importance of carefully tuning the distillation parameters to maximize convergence and generalization, validating the chosen default setting of (temp=1.0, alpha=0.5).

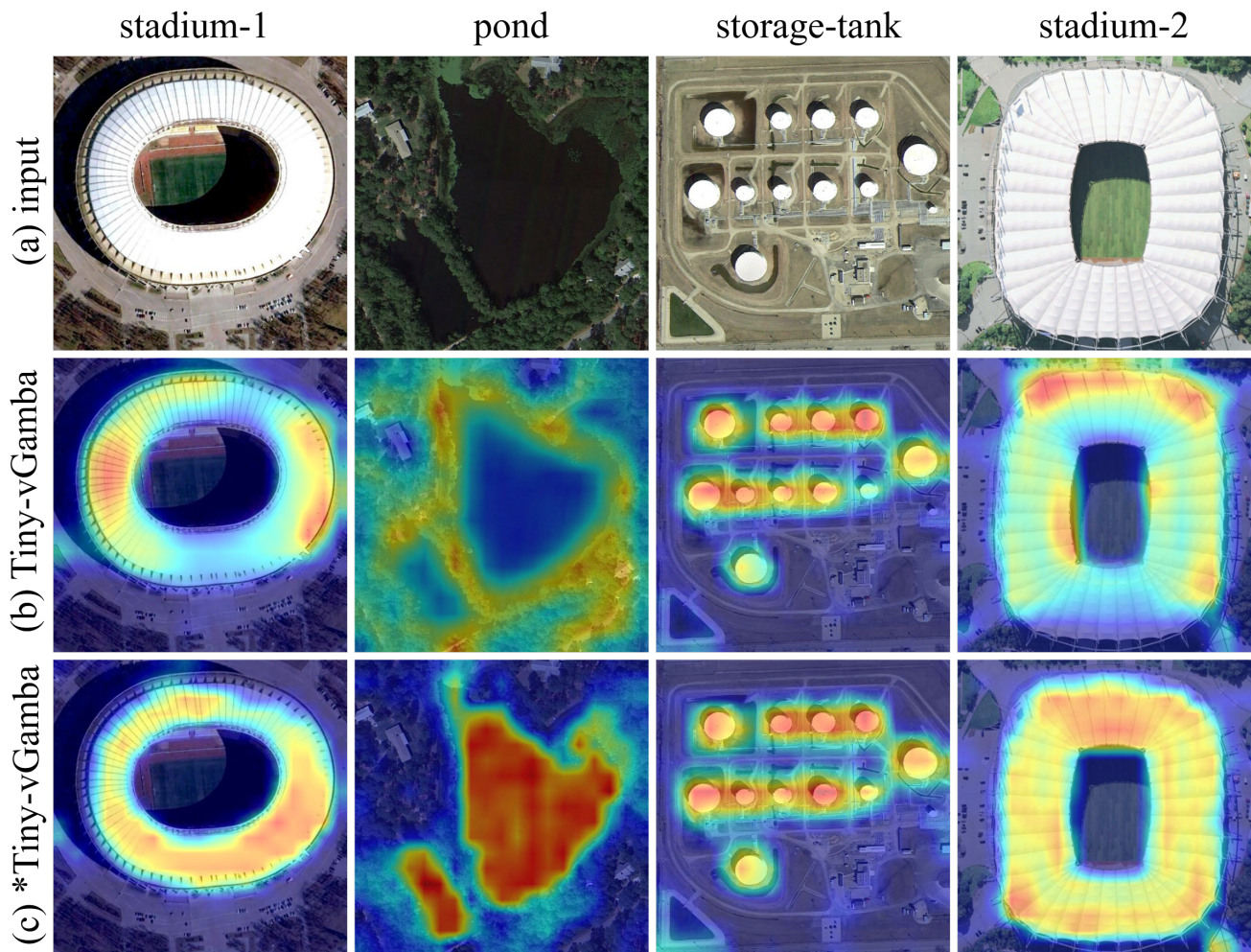


Figure 3. To evaluate the decision-making capabilities of our model, we conducted a qualitative assessment of Tiny-vGamba on the AID dataset. This dataset is particularly challenging due to its reliance on long-range dependencies and the need for capturing global contextual information to make accurate predictions. To demonstrate the improvement in generalization after applying our distillation framework, we employed Hi-ResCAM [3] to visualize heatmaps. The results clearly indicate enhanced attention and decision consistency in the distilled version of Tiny-vGamba.

6. Conclusion

In this work, we proposed Tiny-vGamba, a lightweight visual backbone designed for efficient visual recognition on resource-constrained platforms. By leveraging a hybrid architectural design and integrating knowledge distillation from a frozen CLIP model, we achieved notable performance improvements without increasing the model’s computational complexity. Through quantitative evaluations on IMAGENET1k, COCO, ADE20K dataset and qualitative analysis on AID dataset using HiResCAM, we demonstrated that Tiny-vGamba not only generalizes well but also enhances attention to semantically meaningful regions after distillation. Our results indicate that distilled Tiny-vGamba enhances accuracy by capturing long-range dependencies

while retaining its edge-friendly design, making it suitable for mobile and real-time vision applications.

6.1. Limitations

Tiny-vGamba is currently limited to visual tasks, and its generalization to multi-modal or non-visual domains has not been explored. Further optimization is also necessary for deployment on ultra-low-power edge devices.

6.2. Broader Impact

This research contributes to the development of lightweight visual backbones that efficiently capture long-range dependencies in downstream visual tasks such as classification, segmentation, and object detection. The proposed Tiny-vGamba offer good generalization capabilities and

Table 5. Ablation study on the effects of knowledge distillation loss configurations.

CE Loss	KD Loss	Temp / α	Top-1
✓	✗	– / –	74.2%
✗	✓	1.0 / 0.5	72.8%
✓	✓	0.5 / 0.3	75.4%
✓	✓	1.0 / 0.3	75.9%
✓	✓	2.0 / 0.3	75.2%
✓	✓	0.5 / 0.5	75.8%
✓	✓	1.0 / 0.5	76.7%
✓	✓	2.0 / 0.5	76.0%
✓	✓	0.5 / 0.7	75.6%
✓	✓	1.0 / 0.7	76.2%
✓	✓	2.0 / 0.7	75.4%

well-suited for deployment in resource-constrained environments. By distilling knowledge from large-scale vision-language models into compact architectures like Tiny-vGamba, we also reduce the reliance on large labeled datasets and lowers the computational constrain to training and inference. This work can help make advanced computer vision tools more accessible to areas like healthcare, disaster response, agriculture, and autonomous systems.

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